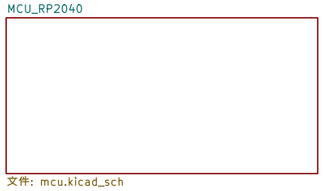
Power



文件: power.kicad_sch

MCU_RP2040



文件: mcu.kicad_sch

RF_1090



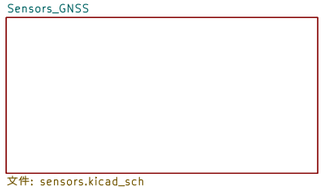
文件: rf1090.kicad_sch

SubGHz_978



文件: subghz.kicad_sch

Sensors_GNSS



文件: sensors.kicad_sch

Interface_J3



文件: interface.kicad_sch

RF_VHF



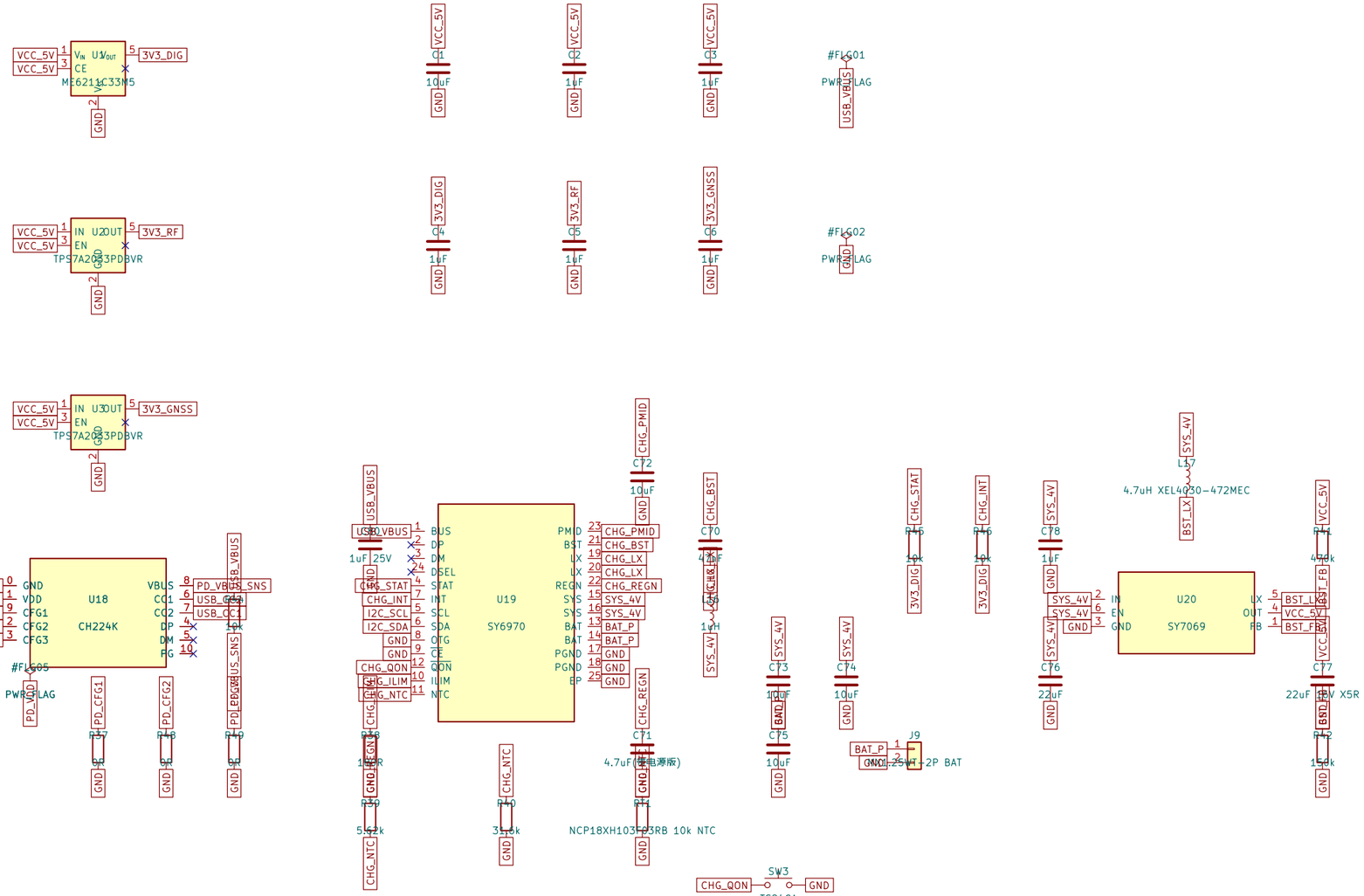
文件: rfvhf.kicad_sch

Topology: independently transcribed, no third-party design files reused
Dual-band ADS-B (1090ES/978UAT) + IMU + BARO + GNSS
Pilot Kit

Sheet: /
File: expansion-board-v4.kicad_sch

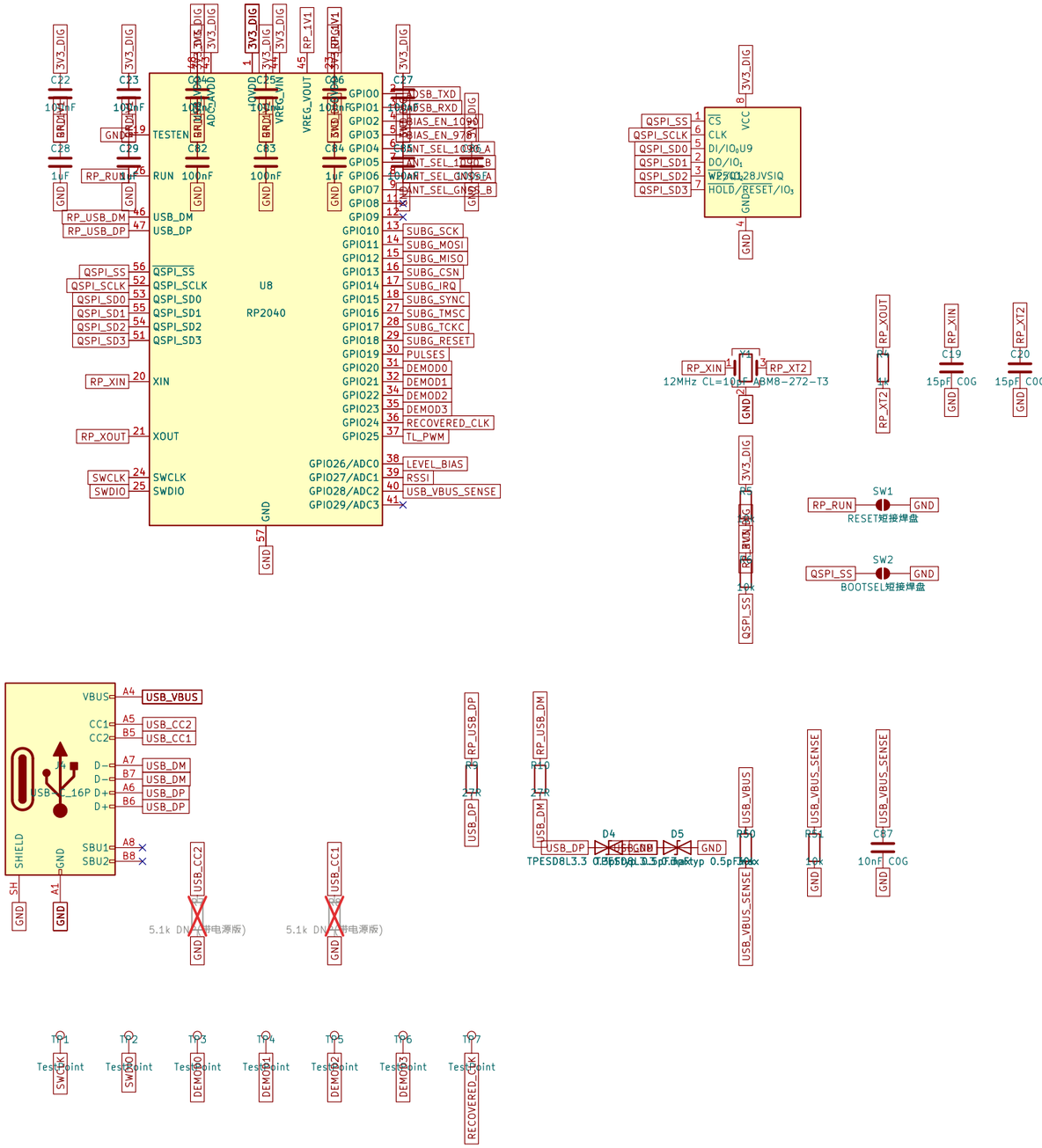
Title: Pilot Kit Expansion Board V4.7 — Root

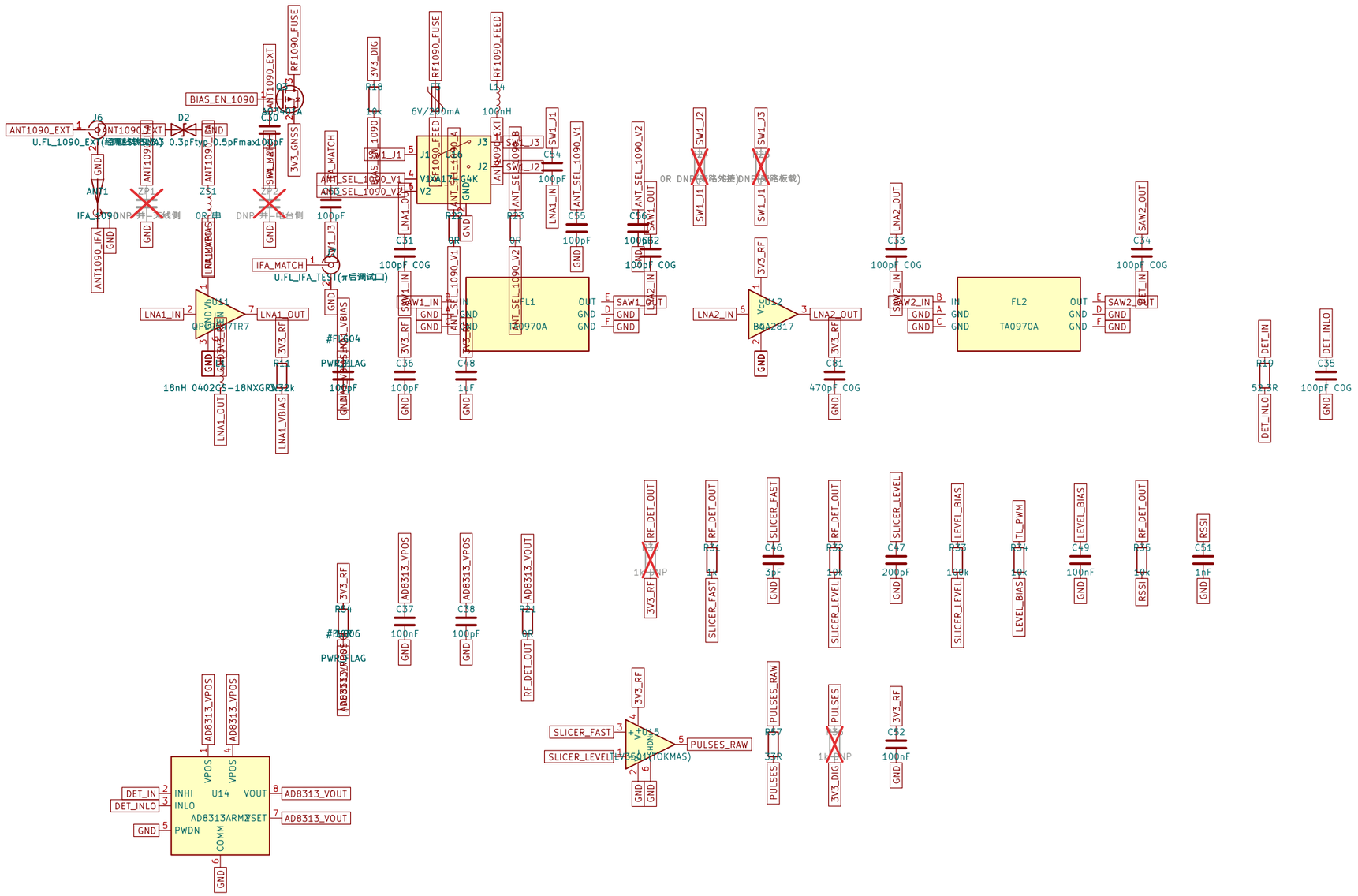
Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 1/8



Pilot Kit

Sheet: /Power/		
File: power.kicad_sch		
Title: Expansion Board V4.7 – Power: 5V in, 3x LDO		
Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 2/8


U8
RP2040
TESTEN
RUN
USB_DM
USB_DP
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
XIN
XOUT
SWCLK
SWDIO
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
SUBG_TXD
SUBG_RXD
SUBG_BIAS_EN_1090
SUBG_BIAS_EN_970
SUBG_ANT_SEL_1090_A
SUBG_ANT_SEL_1090_B
SUBG_ANT_SEL_970_A
SUBG_ANT_SEL_970_B
SUBG_SCK
SUBG_MOSI
SUBG_MISO
SUBG_CSN
SUBG_IRQ
SUBG_SYNC
SUBG_TMSC
SUBG_TCKC
SUBG_RESET
PULSES
DEMOD0
DEMOD1
DEMOD2
DEMOD3
RECOVERED_CLK
TI_PWM
LEVEL_BIAS
RSSI
USB_VBUS_SENSE
3V3_DIG
RP_1V1
GND
C22
C23
C24
C25
C26
C27
C28
C29
C62
C63
C64
C65
C66
C67
C68
C69
C70
C71
C72
C73
C74
C75
C76
C77
C78
C79
C80
C81
C82
C83
C84
C85
C86
C87
1uF
100nF
RP_RUN
RP_USB_DM
RP_USB_DP
RP_XIN
RP_XOUT
RP_XT2
12MHz CL=10pF ABM8-272-T3
15pF C0G
10nF C0G
SW1
SW2
RESET短接焊盘
BOOTSEL短接焊盘
U9
W25Q128JVSIQ
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
R22
R23
R24
R25
R26
R27
R28
R29
R30
R31
R32
R33
R34
R35
R36
R37
R38
R39
R40
R41
R42
R43
R44

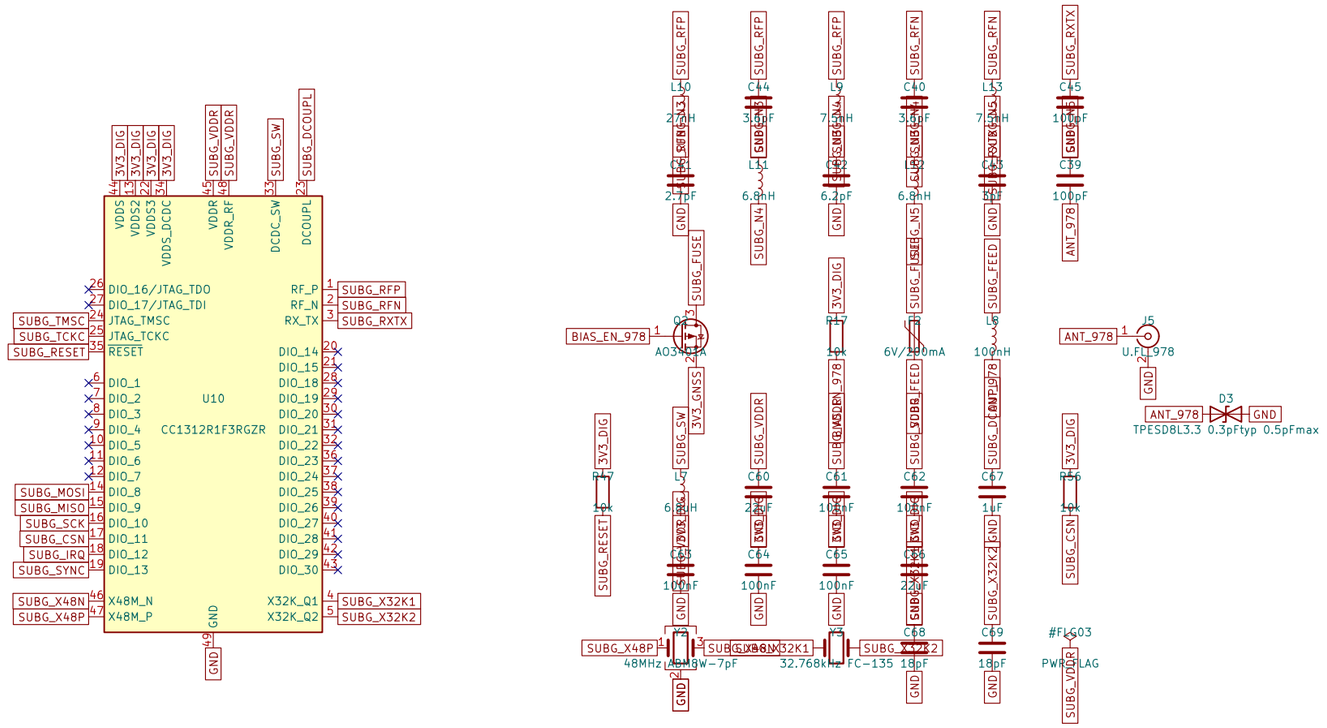


Pilot Kit

Sheet: /RF_1090/
File: rf1090.kicad_sch

Title: Expansion Board V4.7 — 1090MHz: antenna switch + RX chain

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 4/8

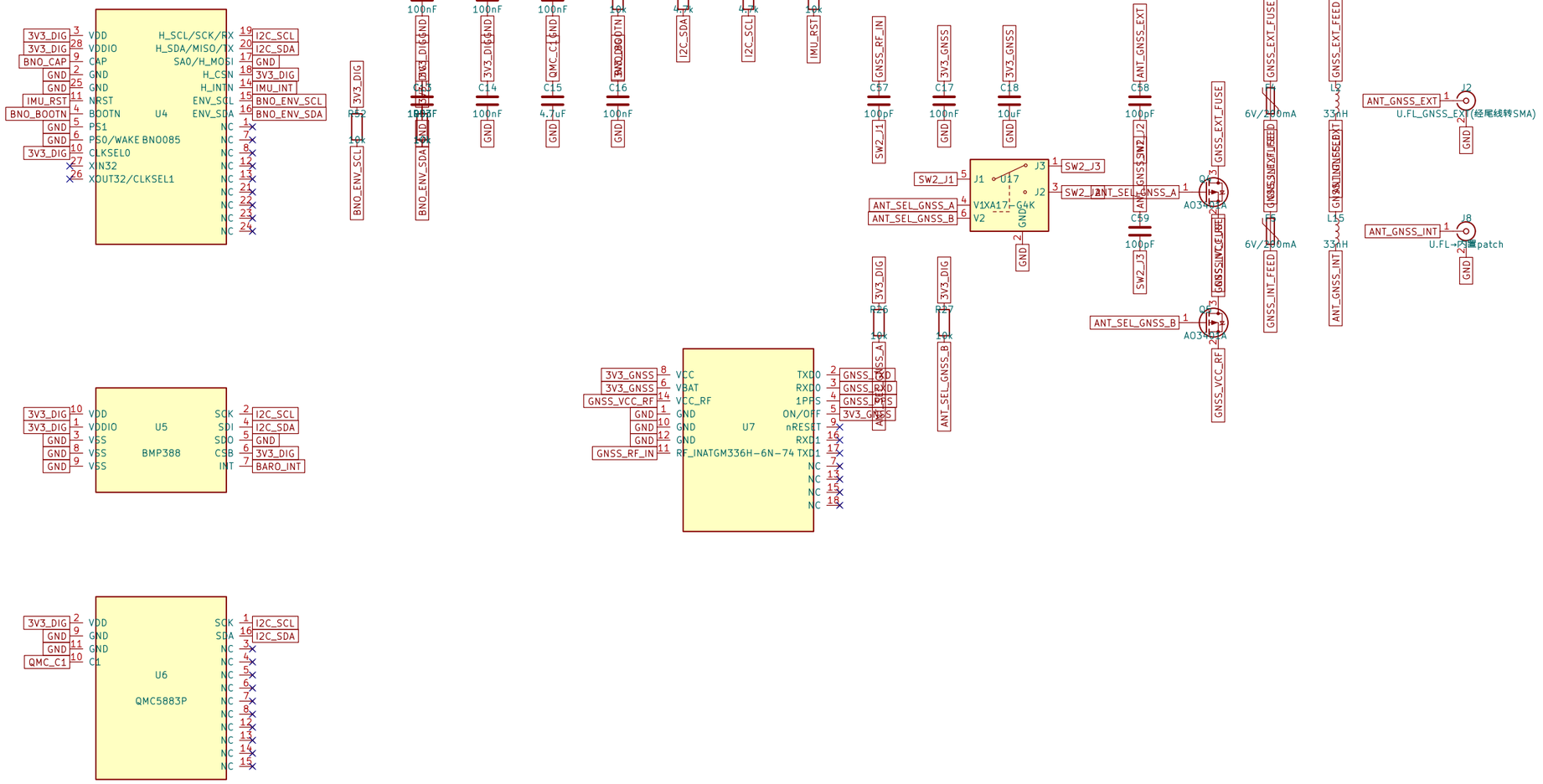


Pilot Kit

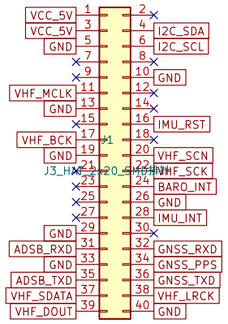
Sheet: /SubGHz_978/
File: subghz.kicad_sch

Title: Expansion Board V4.7 – 978MHz UAT: CC1312R

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 5/8



Pilot Kit		
Sheet: /Sensors_GNSS/		
File: sensors.kicad_sch		
Title: Expansion Board V4.7 – Sensors + GNSS		
Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 6/8

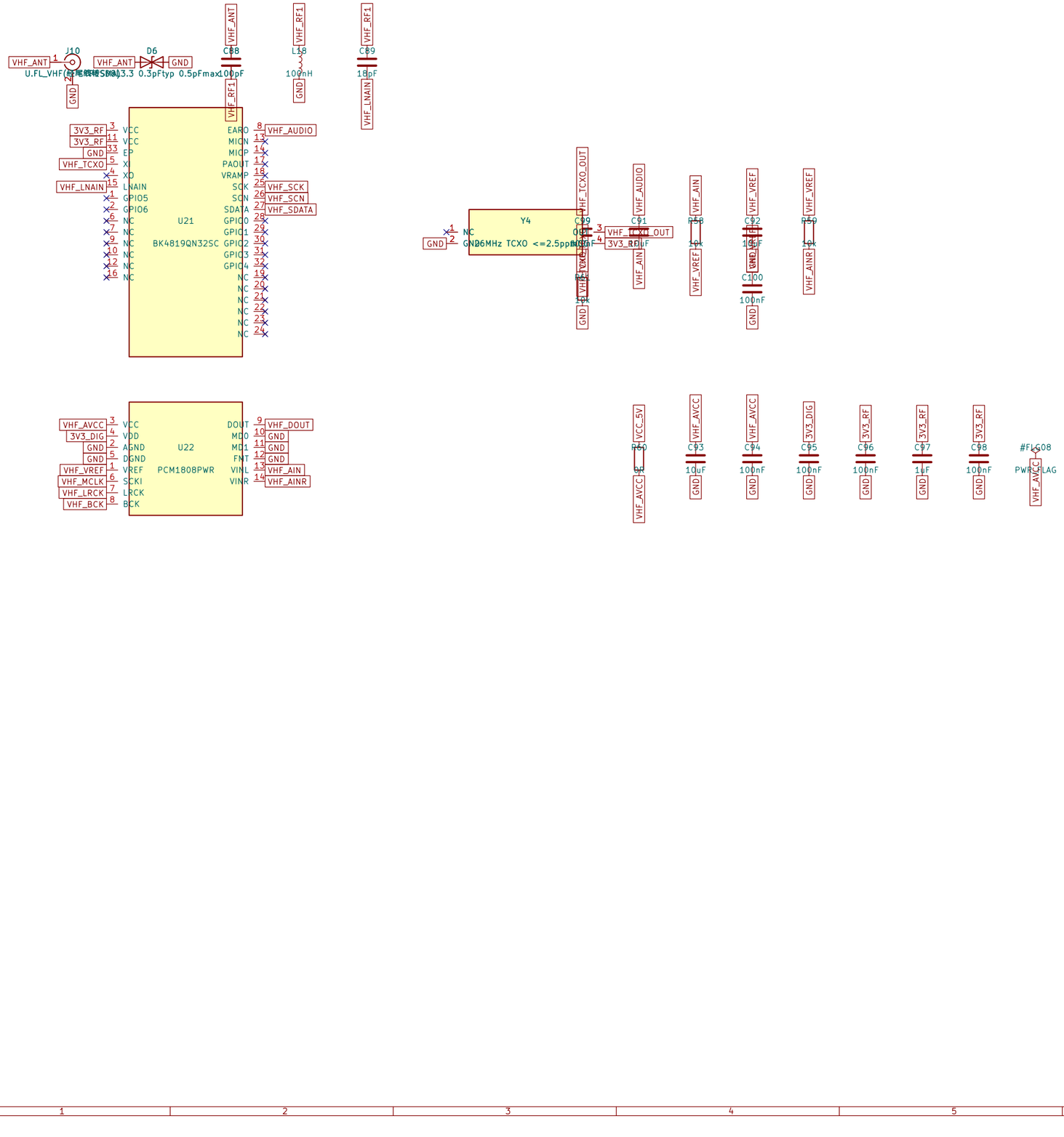


Pilot Kit

Sheet: /Interface_J3/
File: interface.kicad_sch

Title: Expansion Board V4.7 – Interface: J3 2x20 HAT socket

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 7/8



Pilot Kit

Sheet: /RF_VHF/
File: rfvhf.kicad_sch

Title: Expansion Board V4.7 – VHF airband: BK4819 RX + PCM1808 capture

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 8/8